

Complex multiplicative chaos and the critical Ashkin–Teller model

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Based on joint works with Lorca Heeney, Marcin Lis and Avelio Sepúlveda

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FWF Austrian
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The Ashkin–Teller model

For $J \geq 0$, $U \in \mathbb{R}$,

$$\mathbf{P}^{\text{AT}}(\sigma, \tilde{\sigma}) \propto \exp \left\{ \sum_{xy \in E} J\sigma_x\sigma_y + J\tilde{\sigma}_x\tilde{\sigma}_y + U\tau_x\tau_y \right\}, \quad \text{where } \tau_x = \sigma_x\tilde{\sigma}_x.$$

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For $a = e^{2U} \sinh(2J)$, $b = e^{2U} \cosh(2J) - 1$.

$$\mathbf{P}^{\text{RC}}(\omega) \propto 2^{k(\omega)} \sum_{\eta \in \mathcal{E}(\omega)} a^{|\eta|} b^{|\omega \setminus \eta|}.$$

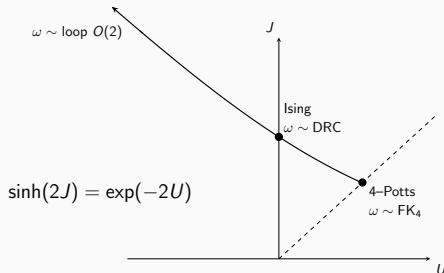
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The Ashkin–Teller model

Proposition [Glazman, Peled 2023 & Lis 2022]:

$$\tau = \sum_{\mathcal{C} \in \mathfrak{C}(\omega)} \xi_{\mathcal{C}} \mu_{\mathcal{C}}, \quad \text{where } \mu_{\mathcal{C}} := \sum_{v \in \mathcal{C}} \delta_v.$$

The critical Ashkin–Teller model

Conjecture [Kadanoff, Brown 1979 & Nienhuis 1984 & ...]:

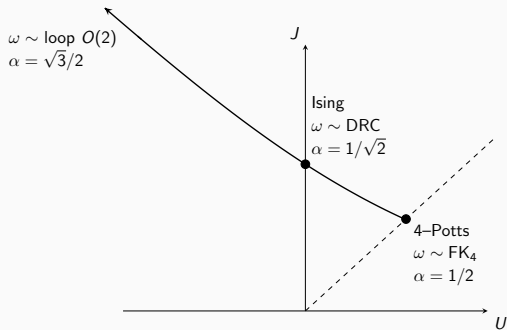
In the scaling limit $\delta \rightarrow 0$,

- Under free boundary conditions, $\delta^{-\alpha^2/2} \tau_\delta \longrightarrow :\sin(\alpha\phi):$
- Under + boundary conditions, $\delta^{-\alpha^2/2} \tau_\delta \longrightarrow :\cos(\alpha\phi):$

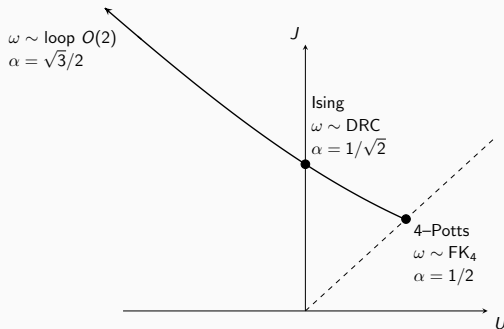
for an explicit $\alpha = \alpha(U) \in [1/2, \sqrt{3}/2) \subset (0, \sqrt{2})$.

(For Juhan $\alpha = 1/\sqrt{g}$)

The critical Ashkin–Teller model



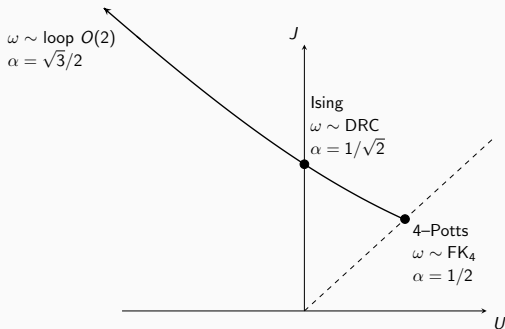
The critical Ashkin–Teller model



Theorem [Junnila, Saksman, Webb 2020 & Bayraktaroglu, Izyurov, Virtanen, Webb 2025]:

Conjecture holds at $\alpha = 1/\sqrt{2}$ in simply connected domains.

The critical Ashkin–Teller model



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Theorem [Chen, Xia 2026]:

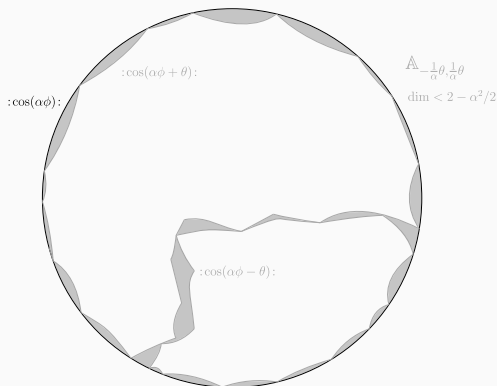
Conjecture holds at $\alpha = 1/2$ in full-plane.

The continuum representation

Rule 1 [Schoug, Sepúlveda, Viklund 2022]:

Explore until hitting $-\frac{1}{\alpha}\theta$ or $\frac{1}{\alpha}\theta$ for $\theta < \pi/2$:

$$:\cos(\alpha\phi): = 0 + \sum_{\ell} :\cos(\alpha\phi^{\ell} \pm \theta):$$

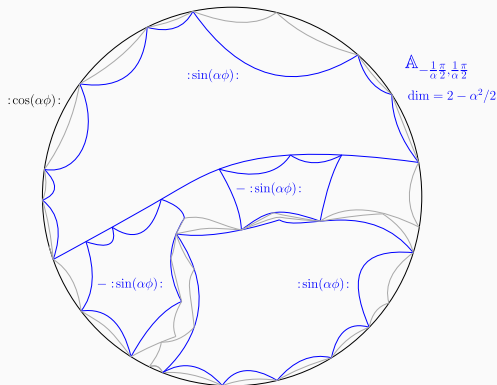


The continuum representation

Rule 2 [AL, Sepúlveda 2026 & Schoug, Sepúlveda, Viklund 2022]:

Explore until hitting $-\frac{1}{\alpha} \frac{\pi}{2}$ or $\frac{1}{\alpha} \frac{\pi}{2}$:

$$:\cos(\alpha\phi): = \mu \mp \sum_{\ell} :\sin(\alpha\phi^{\ell}):$$

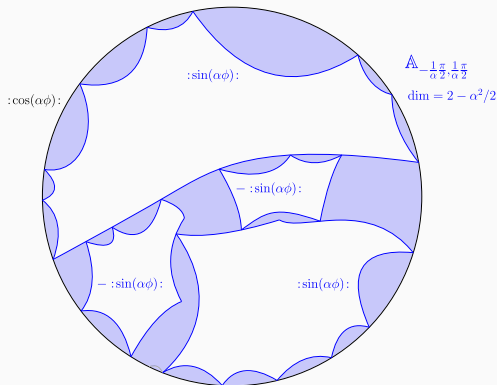


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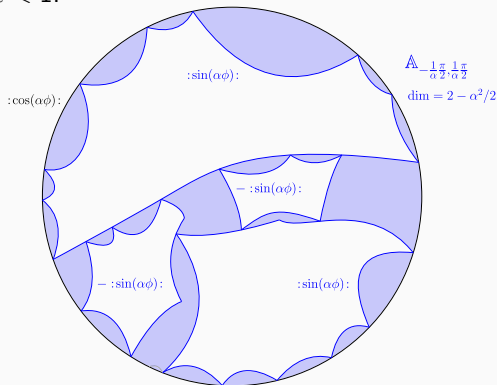
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Note: Forces $\alpha < 1$.

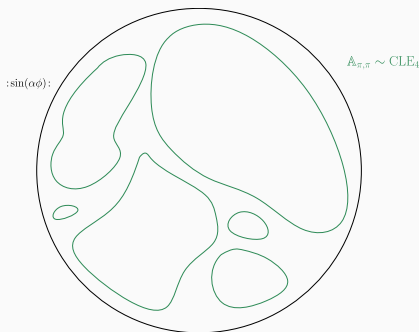


The continuum representation

Rule 3 [AL, Sepúlveda 2026]:

Even if $\alpha \geq 1/2$, explore until hitting $-\pi$ or π :

$$:\sin(\alpha\phi): = 0 + \sum_{\ell} \xi_{\ell} :\cos(\alpha\phi^{\ell} + \xi_{\ell}\pi(\alpha - 1/2)):$$

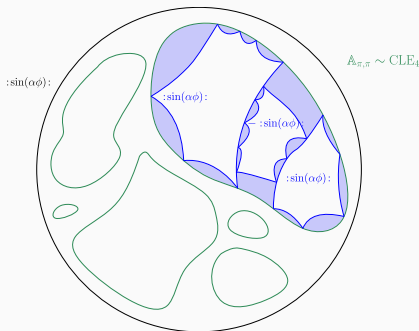


The continuum representation

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$$\begin{aligned} :\sin(\alpha\phi): &= 0 + \sum_{\ell} \xi_{\ell} :\cos(\alpha\phi^{\ell} + \xi_{\ell}\pi(\alpha - 1/2)) : \\ &= 0 + \sum_{\ell} \xi_{\ell} \mu_{\ell} \pm \sum_{\ell} :\sin(\alpha\phi^{\ell}): \end{aligned}$$

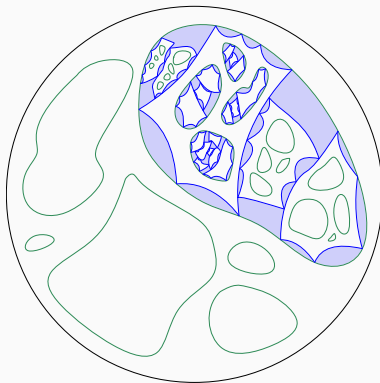


The continuum representation

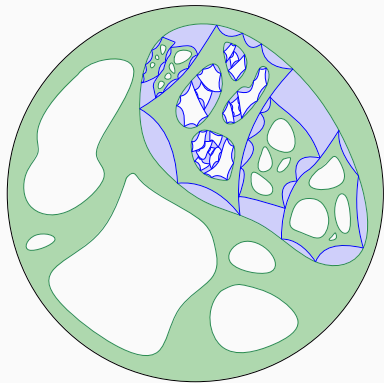
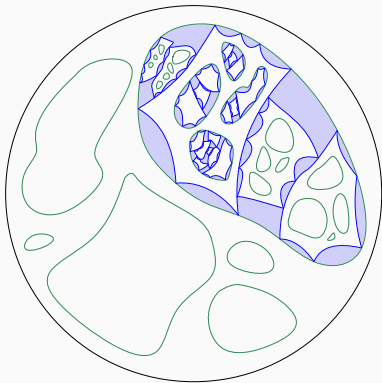
Theorem [AL, Sepúlveda 2026]: For $\alpha \in (0, 1)$,

$$:\sin(\alpha\phi): = \sum_c \xi_c \mu_c,$$

$$:\cos(\alpha\phi): = \mu_0 + \sum_c \xi_c \mu_c.$$



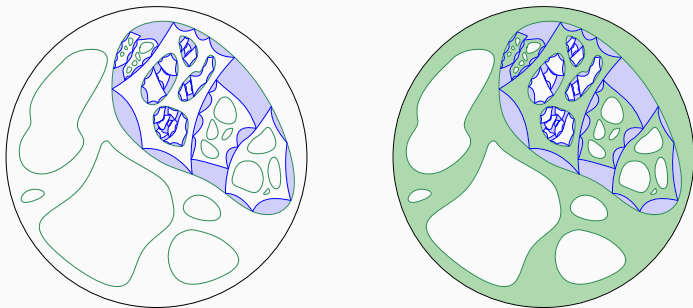
The dual decomposition?



Proposition [AL, Heeney, Lis 2026 & Glazman, Peled 2023 & Lis 2022]:

$$\sigma^* = \sum_{\mathcal{C}^* \in \mathfrak{C}(\omega^*)} \xi_{\mathcal{C}^*} \mu_{\mathcal{C}^*}, \quad \text{where } \mu_{\mathcal{C}^*} := \sum_{v \in \mathcal{C}^*} \delta_v.$$

The single-spin representation



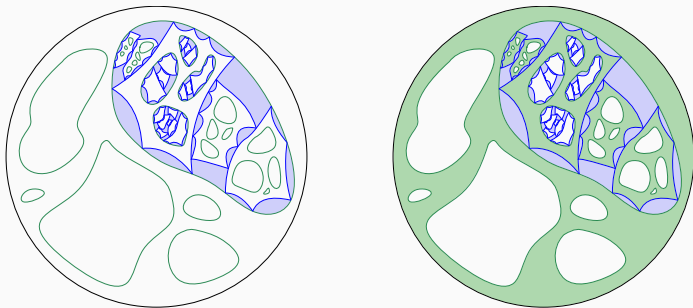
Conjecture [AL, Heeney, Lis 2026 & Aru, Lupu 2026]:

For any U along the critical line, in the scaling limit $\delta \rightarrow 0$,

$$\delta^{-1/8} \sigma_{\delta}^* = \delta^{-1/8} \sum_{\mathcal{C}_{\delta}^* \in \mathfrak{C}(\omega_{\delta}^*)} \xi_{\mathcal{C}_{\delta}^*} \mu_{\mathcal{C}_{\delta}^*} \longrightarrow \sum_{\mathcal{C}^*} \xi_{\mathcal{C}^*} \mu_{\mathcal{C}^*}.$$

Note: The critical exponent is constant along the critical line!

Convergence results for $U = 0$, $\alpha = 1/\sqrt{2}$

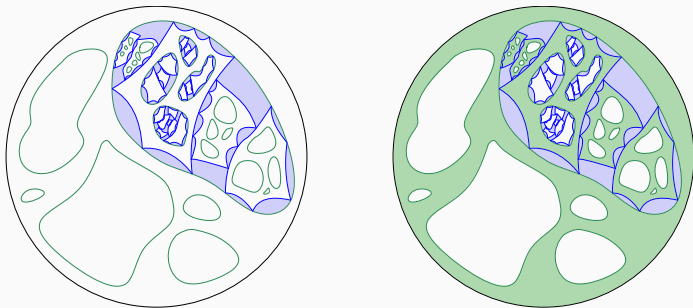


Theorem [Duminil-Copin, Lis, Qian 2025]:

For $U = 0$, the clusters of ω_δ converge to the blue clusters.

Thus the clusters of ω_δ^* converge to the green clusters

Convergence results for $U = 0$, $\alpha = 1/\sqrt{2}$

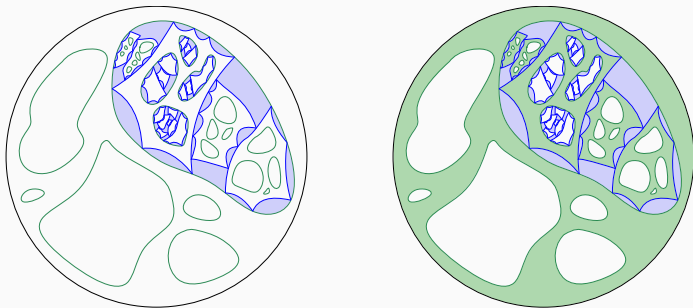


Theorem [AL, Heeney, Lis 2026]:

For $U = 0$ the above conjecture holds.

In particular the Ising model is a measurable function of the GFF and additional coin tosses.

Convergence results for $U = 0$, $\alpha = 1/\sqrt{2}$

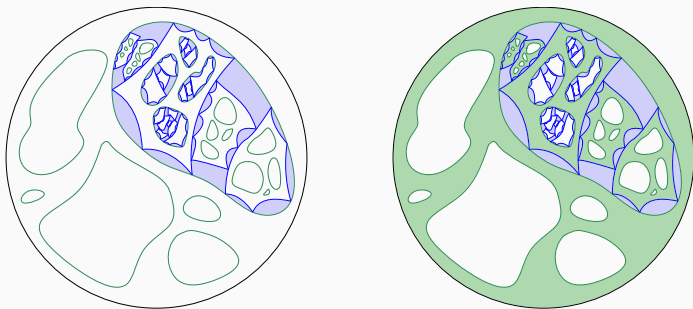


Theorem [Aru, Lupu 2026]:

For $U = 0$,

$$\mathbf{P}(z \overset{c^*}{\longleftrightarrow} w) = \langle \sigma_z \sigma_w \rangle.$$

Convergence results for $U = 0$, $\alpha = 1/\sqrt{2}$



Theorem [AL, Sepúlveda 2026]:

For $U = 0$ the product-spin decomposition also converges, i.e.

$$\delta^{-1/4} \tau_\delta = \delta^{-1/4} \sum_{\mathcal{C}_\delta \in \mathfrak{C}(\omega_\delta)} \xi_{\mathcal{C}_\delta} \mu_{\mathcal{C}_\delta} \longrightarrow \sum_{\mathcal{C}} \xi_{\mathcal{C}} \mu_{\mathcal{C}}.$$

Thank you!

Bonus: self-duality!

In general,

- Different iterations needed to decompose $:\cos(\alpha\phi):$ and $:\sin(\alpha\phi):$.

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But $\alpha = 1/2$ is self-dual: nested CLE_4 .

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But $\alpha = 1/2$ is self-dual: nested CLE_4 .

Work in progress with Juhan Aru: One can (hopefully) compute the CLE_4 two-point function from the complex chaos!

Thank you!